

- [14] B. Chen, F. Xia, B. Ichter, K. Rao, K. Gopalakrishnan, M. S. Ryoo, A. Stone, and D. Kappler. Open-vocabulary queryable scene representations for real world planning. In *2023 IEEE International Conference on Robotics and Automation (ICRA)*, pages 11509–11522. IEEE, 2023.
- [15] J. McCormac, A. Handa, A. Davison, and S. Leutenegger. Semanticfusion: Dense 3d semantic mapping with convolutional neural networks. In *2017 IEEE International Conference on Robotics and automation (ICRA)*, pages 4628–4635. IEEE, 2017.
- [16] M. Runz, M. Buffier, and L. Agapito. Maskfusion: Real-time recognition, tracking and reconstruction of multiple moving objects. In *2018 IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*, pages 10–20. IEEE, 2018.
- [17] R. Dube, A. Cramariuc, D. Dugas, J. Nieto, R. Siegwart, and C. Cadena. SegMap: 3d segment mapping using data-driven descriptors. In *Robotics: Science and Systems (RSS)*, 2018.
- [18] M. Grinvald, F. Furrer, T. Novkovic, J. J. Chung, C. Cadena, R. Siegwart, and J. Nieto. Volumetric Instance-Aware Semantic Mapping and 3D Object Discovery, July 2019. ISSN 2377-3766.
- [19] A. Rosinol, M. Abate, Y. Chang, and L. Carlone. Kimera: an open-source library for real-time metric-semantic localization and mapping, 2020. URL <https://github.com/MIT-SPARK/Kimera>.
- [20] I. Armeni, Z.-Y. He, J. Gwak, A. R. Zamir, M. Fischer, J. Malik, and S. Savarese. 3d scene graph: A structure for unified semantics, 3d space, and camera. In *Proceedings of the IEEE International Conference on Computer Vision*, 2019.
- [21] A. Rosinol, A. Gupta, M. Abate, J. Shi, and L. Carlone. 3D dynamic scene graphs: Actionable spatial perception with places, objects, and humans. In *Robotics: Science and Systems (RSS)*, 2020.
- [22] N. Hughes, Y. Chang, and L. Carlone. Hydra: A real-time spatial perception system for 3D scene graph construction and optimization. In *Robotics: Science and Systems (RSS)*, 2022.
- [23] S. Peng, K. Genova, C. Jiang, A. Tagliasacchi, M. Pollefeys, T. Funkhouser, et al. Openscene: 3d scene understanding with open vocabularies. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 815–824, 2023.
- [24] K. M. Jatavallabhula, A. Kuwajerwala, Q. Gu, M. Omama, T. Chen, S. Li, G. Iyer, S. Saryazdi, N. Keetha, A. Tewari, et al. Conceptfusion: Open-set multimodal 3d mapping. *arXiv preprint arXiv:2302.07241*, 2023.
- [25] A. Takmaz, E. Fedele, R. W. Sumner, M. Pollefeys, F. Tombari, and F. Engelmann. Open-Mask3D: Open-Vocabulary 3D Instance Segmentation. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.
- [26] S. Lu, H. Chang, E. P. Jing, A. Boularias, and K. Bekris. Ovir-3d: Open-vocabulary 3d instance retrieval without training on 3d data. In *Conference on Robot Learning*, pages 1610–1620. PMLR, 2023.
- [27] Q. Gu, A. Kuwajerwala, S. Morin, K. M. Jatavallabhula, B. Sen, A. Agarwal, C. Rivera, W. Paul, K. Ellis, R. Chellappa, et al. Conceptgraphs: Open-vocabulary 3d scene graphs for perception and planning. *arXiv preprint arXiv:2309.16650*, 2023.
- [28] A. Werby, C. Huang, M. Büchner, A. Valada, and W. Burgard. Hierarchical open-vocabulary 3d scene graphs for language-grounded robot navigation. *Robotics: Science and Systems*, 2024.
- [29] D. Maggio, Y. Chang, N. Hughes, M. Trang, D. Griffith, C. Dougherty, E. Cristofalo, L. Schmid, and L. Carlone. Clío: Real-time task-driven open-set 3d scene graphs, 2024.

- [30] A. Pal, Y. Qiu, and H. Christensen. Learning hierarchical relationships for object-goal navigation. In *Conference on Robot Learning*, pages 517–528. PMLR, 2021.
- [31] J. Ye, D. Batra, A. Das, and E. Wijmans. Auxiliary tasks and exploration enable object goal navigation. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 16117–16126, 2021.
- [32] A. Khandelwal, L. Weihs, R. Mottaghi, and A. Kembhavi. Simple but effective: Clip embeddings for embodied ai. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 14829–14838, 2022.
- [33] Z. Al-Halah, S. K. Ramakrishnan, and K. Grauman. Zero experience required: Plug & play modular transfer learning for semantic visual navigation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 17031–17041, 2022.
- [34] A. Majumdar, G. Aggarwal, B. Devnani, J. Hoffman, and D. Batra. Zson: Zero-shot object-goal navigation using multimodal goal embeddings. *Advances in Neural Information Processing Systems*, 35:32340–32352, 2022.
- [35] U. Jain, I.-J. Liu, S. Lazebnik, A. Kembhavi, L. Weihs, and A. G. Schwing. Gridtopix: Training embodied agents with minimal supervision. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 15141–15151, 2021.
- [36] H. Luo, A. Yue, Z.-W. Hong, and P. Agrawal. Stubborn: A strong baseline for indoor object navigation. In *2022 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 3287–3293. IEEE, 2022.
- [37] G. Georgakis, B. Bucher, K. Schmeckpeper, S. Singh, and K. Daniilidis. Learning to map for active semantic goal navigation. *arXiv preprint arXiv:2106.15648*, 2021.
- [38] M. Zhu, B. Zhao, and T. Kong. Navigating to objects in unseen environments by distance prediction. In *2022 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 10571–10578. IEEE, 2022.
- [39] S. Y. Gadre, M. Wortsman, G. Ilharco, L. Schmidt, and S. Song. Cows on pasture: Baselines and benchmarks for language-driven zero-shot object navigation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 23171–23181, 2023.
- [40] K. Zhou, K. Zheng, C. Pryor, Y. Shen, H. Jin, L. Getoor, and X. E. Wang. Esc: Exploration with soft commonsense constraints for zero-shot object navigation. In *International Conference on Machine Learning*, pages 42829–42842. PMLR, 2023.
- [41] A. Staroverov, K. Muravyev, K. Yakovlev, and A. I. Panov. Skill fusion in hybrid robotic framework for visual object goal navigation. *Robotics*, 12(4):104, 2023.
- [42] J. Chen, G. Li, S. Kumar, B. Ghanem, and F. Yu. How to not train your dragon: Training-free embodied object goal navigation with semantic frontiers. In *Robotics: Science and Systems (RSS)*, 2023. doi:10.15607/RSS.2023.XIX.075.
- [43] D. Shah, M. Equi, B. Osinski, F. Xia, B. Ichter, and S. Levine. Navigation with large language models: Semantic guesswork as a heuristic for planning. *arXiv preprint arXiv:2310.10103*, 2023.
- [44] V. S. Dorbala, J. F. Mullen, and D. Manocha. Can an embodied agent find your “cat-shaped mug”? Llm-based zero-shot object navigation. *IEEE Robotics and Automation Letters*, 9(5): 4083–4090, 2024. doi:10.1109/LRA.2023.3346800.

- [45] T.-Y. Lin, M. Maire, S. Belongie, J. Hays, P. Perona, D. Ramanan, P. Dollár, and C. L. Zitnick. Microsoft coco: Common objects in context. In *Computer Vision—ECCV 2014: 13th European Conference, Zurich, Switzerland, September 6-12, 2014, Proceedings, Part V 13*, pages 740–755. Springer, 2014.
- [46] A. Kuznetsova, H. Rom, N. Alldrin, J. Uijlings, I. Krasin, J. Pont-Tuset, S. Kamali, S. Popov, M. Mallocci, A. Kolesnikov, et al. The open images dataset v4: Unified image classification, object detection, and visual relationship detection at scale. *International Journal of Computer Vision*, 128(7):1956–1981, 2020.
- [47] G. Cloud. Vision ai. URL <https://cloud.google.com/vision>.
- [48] M. A. A. Services. Image tagging. URL <https://learn.microsoft.com/en-us/azure/ai-services/computer-vision/concept-tagging-images>.
- [49] A. Developer. Classifying images for categorization and search. URL https://developer.apple.com/documentation/vision/classifying_images_for_categorization_and_search.
- [50] Y. Zhang, X. Huang, J. Ma, Z. Li, Z. Luo, Y. Xie, Y. Qin, T. Luo, Y. Li, S. Liu, et al. Recognize anything: A strong image tagging model. *arXiv preprint arXiv:2306.03514*, 2023.
- [51] X. Huang, Y.-J. Huang, Y. Zhang, W. Tian, R. Feng, Y. Zhang, Y. Xie, Y. Li, and L. Zhang. Open-set image tagging with multi-grained text supervision. *arXiv e-prints*, pages arXiv–2310, 2023.
- [52] K. N. Kutulakos and S. M. Seitz. A theory of shape by space carving. *International Journal of Computer Vision*, 38:199–218, 2000.
- [53] M. Ester, H.-P. Kriegel, J. Sander, and X. Xu. A density-based algorithm for discovering clusters in large spatial databases with noise. In *Proceedings of the Second International Conference on Knowledge Discovery and Data Mining, KDD’96*, page 226–231. AAAI Press, 1996.
- [54] A. A. Taha and A. Hanbury. Metrics for evaluating 3d medical image segmentation: analysis, selection, and tool. *BMC medical imaging*, 15(1):1–28, 2015.
- [55] A. Chang, A. Dai, T. Funkhouser, M. Halber, M. Niessner, M. Savva, S. Song, A. Zeng, and Y. Zhang. Matterport3D: Learning from RGB-D data in indoor environments. *International Conference on 3D Vision (3DV)*, 2017.
- [56] M. Hutter, C. Gehring, D. Jud, A. Lauber, C. D. Bellicoso, V. Tsounis, J. Hwangbo, K. Bodie, P. Fankhauser, M. Bloesch, et al. Anymal-a highly mobile and dynamic quadrupedal robot. In *2016 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 38–44. IEEE, 2016.
- [57] J. L. Schönberger and J.-M. Frahm. Structure-from-motion revisited. In *Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016.
- [58] J. L. Schönberger, E. Zheng, M. Pollefeys, and J.-M. Frahm. Pixelwise view selection for unstructured multi-view stereo. In *European Conference on Computer Vision (ECCV)*, 2016.
- [59] C. Cao, H. Zhu, F. Yang, Y. Xia, H. Choset, J. Oh, and J. Zhang. Autonomous exploration development environment and the planning algorithms. In *2022 International Conference on Robotics and Automation (ICRA)*, pages 8921–8928. IEEE, 2022.
- [60] J. Achiam, S. Adler, S. Agarwal, L. Ahmad, I. Akkaya, F. L. Aleman, D. Almeida, J. Altenschmidt, S. Altman, S. Anadkat, et al. Gpt-4 technical report. *arXiv preprint arXiv:2303.08774*, 2023.